

Mohammad Aria Amini

Geology Graduate

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ABOUT

I am Mohammad Aria Amini, a Bachelor's graduate in Geology and a researcher, writer, and scientific editor in the field of Earth sciences. I am interested in research, scientific writing and editing, and continuously developing my knowledge and professional skills in Earth sciences.

INTERESTS

My research and professional interests primarily lie in the field of Earth sciences, with a particular focus on geochemistry, petroleum geology, and topics related to subsurface resources. I am interested in studying various areas of geology, particularly geological and geochemical processes, rocks and minerals, petroleum systems, and the application of geological data to the understanding and assessment of subsurface resources. I am also interested in applying modern data processing, analytical methods, and computational technologies to Earth science research, and I continuously seek to expand my knowledge and skills in these areas.

EDUCATION

B.Sc. in Geology 2022 – 2026

Golestan University, Faculty of Earth Sciences, Gorgan, Iran

- Ranked first among incoming undergraduate students, with a Grade A average
- Completed the program in seven semesters – one semester ahead of the standard curriculum
- Completed 6 of 7 terms at Shahid Beheshti University, Tehran

High School Diploma, Experimental Sciences 2017 – 2020

Shahid Beheshti Exceptional Talents High School (NODET), Ahvaz, Iran

PROFESSIONAL EXPERIENCE

Varjavand Intelligent Creative Software & Hardware Services Co.

Chief Executive Officer (CEO) and Chairman of the Board · Full-Time

Website: varjavand.org

Jul 2025 – Present

Arian Zamin Publishing House

Director · Full-Time

Apr 2026 – Present

Editorial Specialist and Publications Researcher · Part-Time

Mar 2023 – Apr 2026

Pars Geological Research Center (Arian Zamin)

Geology Specialist · Full-Time

Feb 2023 – Present

Gemology Science and Art Development Co. (Gemsa), Knowledge-Based Company

Research & Development (R&D) Manager · Full-Time

Sep 2024 – Apr 2025

RESEARCH & INVESTIGATIVE EXPERIENCE

Academic Internship

National Iranian Oil Company

Internship report: National Iranian South Oilfields Company, Geological Operations Department · Grade A

May 2025 – Aug 2025

Research & Development

Gemology Science and Art Development Co. (Gemsa)

Research on Raman spectroscopy using spectrophotometric methods · Supervised by Prof. Fariborz Masoudi (Full Professor)

Sep 2024 – Apr 2025

Research Collaborator and Manager of HPC Infrastructure

Institute of Medicinal Plants and Raw Materials, Shahid Beheshti University

Designed, built, and operated the group's computing infrastructure and processing environment · Directly assisted with running and compiling code, and supported its users · Supervised by Dr. Hassan Rezadoost Chahardehi (Associate Professor) · h_rezadoost@sbu.ac.ir

Oct 2024 – Present

Research Collaborator and Manager, Physical Chemistry Computing (HPC) Core

Faculty of Chemistry and Petroleum, Shahid Beheshti University, Tehran

Managing and developing the faculty's HPC infrastructure to support computational and physical chemistry research · Also assisted with managing computing users, compiling code, and running computations · Supervised by Prof. Mansour Zahedi (Full Professor) · m-zahedi@sbu.ac.ir

Apr 2025 – Present

Earth Sciences and Petroleum Researcher

Pars Geological Research Center (Arian Zamin)

Research on economic geology, mineral deposits, and petroleum geology, including petrographic and geochemical data analysis · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Jan 2023 – Present

TEACHING ACTIVITIES

Teaching Assistant (TA) — Mining Engineering Technical Services and Mining Regulations

Feb 2026 – Sep 2026

Shahid Beheshti University, Tehran, Faculty of Earth Sciences

Course for 8th-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Teaching Assistant (TA) — Igneous Petrology

Oct 2025 – Feb 2026

Shahid Beheshti University, Tehran, Faculty of Earth Sciences

Course for third-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Teaching Assistant (TA) — Igneous Petrology

Oct 2024 – Feb 2025

Shahid Beheshti University, Tehran, Faculty of Earth Sciences

Course for third-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Research Week Events & Workshops Series

Jan 2026 – Feb 2026

Shahid Beheshti University, Tehran

Instructor · Workshop: Introduction to Parallel Processing Systems and Linux-Based High-Performance Computing Environments

PATENTS

System and Method for Depth-Aware Identification and Zonation of Productive (Net Pay) Intervals in Oil Wells Using Multi-Model Probabilistic Fusion of Well-Log Data

Registered · 2026

Iranian Patent · Owner and Sole Inventor

Application No. 140450140003008805 · Filed 18 Dec 2025 (1404/09/27) · Registration No. 114471 · Published 23 Sep 2026 (1405/07/01) · IPC: E21B 43/00 ·

[View on Iranian Patent Office](#) · [View Patent Registration Certificate](#)

A depth-aware software system that standardizes and quality-controls multi-well log data, builds depth-based features, trains several classifiers, calibrates their probabilities and fuses them into per-depth zone labels with an uncertainty index.

BOOKS

Ghorbani, M., & Amini, M. A. (2025). The Historiography of Earth Science in Iran: Introducing a Number of Contemporary Pioneers.

Co-author · ISBN: 978-600-6058-43-6

Fathi, T. (2026). Petroleum Contamination Hydrogeology: Geological, Environmental, Legal Aspects, and Models for the Remediation of Contaminated Groundwater.

Scientific Editor · ISBN (from the second edition): 978-600-6058-60-3

Momenzadeh, M. (2025). The Mythical City of Zabol: The Burnt City.

Editor · ISBN: 978-600-6058-44-3

Salmati, R. (2025). A Comprehensive Guide to Geological Mapping at Scales of 1:50,000 and Larger.

Editor · ISBN: 978-600-6058-42-9

CONFERENCES

Paper Title: Natural Resources (Mineral, Energy, and Water Resources) in Iran

2025

4th Global Summit on Advances in Earth Science & Climate Change (Adv. ESCC 2025) - Berlin, Germany (Virtual Presentation)

Oral Presentation

Authors: Mansour Ghorbani, Mohammad Aria Amini (Speaker) · Distinguished Speaker · Organized by Peers Alley Media · Sep 29–30, 2025

Paper Title: Differences in Garnet Classification Based on Color

2024

43rd Earth Sciences Congress (National Conference) - Tehran, Iran

Conference Paper

Authors: Ziba Delpasand, Fariborz Masoudi, Mohammad Aria Amini · 10 pages · Persian · National Sci-Doc ID: GSI43_151 · [View on Civilica](#)

Paper Title: Petrographic, Mineralization, and Alteration Study of the Choudarchay Copper Vein Deposit

2024

43rd Earth Sciences Congress (National Conference) - Tehran, Iran

Oral Presentation

Author: Mohammad Aria Amini · 13 pages · Persian · National Sci-Doc ID: GSI43_064 · [View on Civilica](#)

Paper Title: Economic Geology of Copper and Associated Elements in the South Rayen Area, with a Focus on Gadar-e-Siah

2023

42nd Earth Sciences Congress (National Conference) - Tehran, Iran

Oral Presentation

Authors: Mohammad Aria Amini, Shohreh Salkhordeh · 15 pages · Persian · National Sci-Doc ID: GSI42_161 · [View on Civilica](#)

HONORS & AWARDS

Top Young Earth Sciences Researcher

2024

Geological Society of Iran

Awarded for the paper "Petrographic, Mineralization, and Alteration Study of the Choudarchay Copper Vein Deposit"

LANGUAGES

Persian – Native | English – Professional Working Proficiency | French – Learning

GRE General Test Taken: Dec 11, 2024 · Expires: Dec 11, 2029 · Verbal Reasoning 166/170 · Quantitative Reasoning 166/170 · Analytical Writing 4.0/6.0 (63rd percentile)

TOEFL iBT Taken: Jan 1, 2025 · Expires: Jan 1, 2027 · Total Score 102/120 · Reading 30/30 · Listening 27/30 · Writing 28/30 · Speaking 17/30

PROFESSIONAL MEMBERSHIPS

Member of the Iranian Geological Society

Oct 2022 – Present

Full (regular) member of the society, and also Executive Secretary and Website Manager of the Iranian Geological Society

Member of the Geological Society of America (GSA)

Member of the American Association of Petroleum Geologists (AAPG)

VOLUNTARY & SOCIAL ACTIVITIES

Volunteer Supporting Mahak Hospital and Rehabilitation Complex & Patients <i>MAHAK Charity — The Society to Support Children Suffering from Cancer</i> Disaster and Humanitarian Relief · MAHAK ID: 020175959 · MAHAK Membership No.: 10704	Mar 2023 – Present
Organ Donation Ambassador <i>Iranian Society of Organ Donation</i> Health	Apr 2023 – Present
Organ Donation Volunteer <i>Iranian Society of Organ Donation</i> Disaster and Humanitarian Relief · Membership No.: 3751462 (ehda.ir)	Aug 2019 – Present
Volunteer <i>Iranian Red Crescent Society Innovation Center</i> Disaster and Humanitarian Relief	Apr 2023 – Present
Benevolent Member <i>Raad Charity</i> Veteran Support & Physical Disability Support · Membership No.: 421297 Raad Charity is a private charitable organization dedicated to supporting and empowering people with physical disabilities.	Apr 2023 – Present

TECHNICAL & SPECIALIZED SKILLS

Earth Sciences & Petroleum Geology

Petroleum Geology · Geochemistry · Petrography & Thin-Section Analysis · Geological Mapping · Geological Data Interpretation · Sedimentology & Stratigraphy · Structural Geology & Tectonics

Geoscience-Specific Software & Tools

ArcGIS / ArcGIS Pro · QGIS · Google Earth Engine · ENVI · RockWorks · Petrel

Programming, Computing & Data Analysis

Python · MATLAB · C# · JavaScript · PHP · Bash / Shell Scripting · Data Analysis & Visualization · Geoscientific Data Modeling & Analysis · Large-Scale Data Processing · High-Performance Computing (HPC)

Infrastructure, Systems & Networking

Linux · Linux System Administration · Server Administration · Computer Networking · Network Infrastructure Management · Virtualization & Virtual Machine Management · Data Center Infrastructure · Server & Compute Resource Management

General Software & Productivity Tools

Microsoft Office (Word, Excel, PowerPoint) · Git · GitHub · Command-Line Environments · Technical Documentation

CERTIFICATES, LICENSES & COURSES

Certificates

Vocational Skill Certificate — WordPress CMS Developer · Technical and Vocational Training Organization of Iran · Standard ID 7321205300201 · 2025 · 42245009 · [View certificate](#)

Vocational Skill Certificate — Goldsmith · Technical and Vocational Training Organization of Iran · Standard ID 7313207900301 · 2025 · 42788555 · [View certificate](#)

Petroleum Engineering with AI Applications · Coursera Specialization · [View Specialization Certificate](#)

Introduction to Environmental Science · Coursera Specialization · [View Specialization Certificate](#)

Google Digital Marketing & E-commerce Professional Certificate · Coursera Professional Certificate · [View Professional Certificate](#)

Licenses

Varjavand Intelligent Creative Software Co. — Activity License · Tehran Provincial Computer Trade Organization · [View license](#)

Home Business License — Artificial Intelligence · [View license](#)

Home Business License — Software System Architecture Design · [View license](#)

Home Business License — Gold & Jewelry Design · [View license](#)

Home Business License — Precious & Semi-Precious Gemstone Cutting · [View license](#)

Home Business License — Software Programming · [View license](#)

Business License — Software Installation & Setup Services and Related Activities (Legal Entity)

Courses

Petroleum Engineering with AI Applications

Coursera Specialization · 5 courses

– Hydrocarbon Exploration and Production · 2025 · EVKUGMGT45RP · [View certificate](#)

– From Wellhead to Refinery: Midstream Oil and Gas Processing · 2025 · RO8QDU44OUOL · [View certificate](#)

– Natural Gas Production and Processing · 2025 · IGS15UQQFRV4 · [View certificate](#)

– AI & ML Applications in Oil and Gas Industry · 2025 · O4Y2QT274ZI3 · [View certificate](#)

– Utilities, Safety & Environmental Care in Oil & Gas Industry · 2025 · UQWIYV2XAVAV · [View certificate](#)

[View Specialization Certificate](#)

Introduction to Environmental Science

Coursera Specialization · 3 courses

– Environmental Science · 2024 · 67L7TVDOHCS4 · [View certificate](#)

– Population, Food, and Soil · 2024 · E1UHZEYRKK2T · [View certificate](#)

– Energy and Environment · 2024 · TAEBBTUG4LV3 · [View certificate](#)

[View Specialization Certificate](#)

Google Digital Marketing & E-commerce Professional Certificate

Coursera Professional Certificate · 7 courses

- Foundations of Digital Marketing and E-commerce 2024 · ADAC17OAAUTM · [View certificate](#)
- Attract and Engage Customers with Digital Marketing 2024 · BYDLT1NC85UY · [View certificate](#)
- From Likes to Leads: Interact with Customers Online 2024 · OY3RTJ0FB1O9 · [View certificate](#)
- Think Outside the Inbox: Email Marketing 2025 · J8KTD6HZ13TU · [View certificate](#)
- Assess for Success: Marketing Analytics and Measurement 2025 · Z4TCNOZKS8MI · [View certificate](#)
- Make the Sale: Build, Launch, and Manage E-commerce Stores 2025 · TOTMC7EZJCGQ · [View certificate](#)
- Satisfaction Guaranteed: Develop Customer Loyalty Online 2025 · 7IHL2H68QMKE · [View certificate](#)

[View Professional Certificate](#)

Sustainable Neighborhoods Johns Hopkins University · 2025 · PD3013GV4N42 · [View certificate](#)

Sustainable Transportation Networks and Streetscapes Johns Hopkins University · 2025 · BQ9NUFWQ8JHO · [View certificate](#)

Wind Energy Technical University of Denmark (DTU) · 2025 · BF3120EB7YPC · [View certificate](#)

Sustainable Regional Principles, Planning and Transportation Johns Hopkins University · 2025 · HIHAH8WSFR95 · [View certificate](#)

Our Earth: Its Climate, History, and Processes University of Manchester · 2023 · 6Y537YJ5QH9J · [View certificate](#)

Extinctions: Past, Present, & Future Emory University · 2023 · EK5WJUBLEB92 · [View certificate](#)

Electric Power Systems University at Buffalo · 2023 · QFFR5N2AQ26W · [View certificate](#)

Introduction to GIS Mapping University of Toronto · 2023 · RXDSMTEBNYCY · [View certificate](#)

Solar Energy Basics The State University of New York · 2023 · PTMSZ9B6KQ8F · [View certificate](#)

Oil & Gas Industry Operations and Markets Duke University · 2023 · G7DYSPR7M3RR · [View certificate](#)

Unity: Design & Deform Meshes for 3D Geometry Control EDUCBA · 2025 · PAOC4O9D4ALK · [View certificate](#)

Positive Psychiatry and Mental Health The University of Sydney · 2025 · LQ39UU674WEF · [View certificate](#)

Firm Level Economics: Markets and Allocations University of Illinois Urbana-Champaign · 2024 · XICPQ5ZM9KSV · [View certificate](#)

Introduction to Web Development University of California, Davis · 2024 · OISIZ0VHN9AP · [View certificate](#)

How to Write and Publish a Scientific Paper École Polytechnique · 2024 · TRG7BMSS8Q9I · [View certificate](#)

Firm Level Economics: Consumer and Producer Behavior University of Illinois Urbana-Champaign · 2024 · 1JH025ZXSAFZ · [View certificate](#)

Build a Full Website using WordPress Coursera · 2023 · FLH3PRW9CBDC · [View certificate](#)

COVID-19 Contact Tracing Johns Hopkins University · 2023 · CTXPE9DBBSWB · [View certificate](#)

High Stakes Leadership: Leading in Times of Crisis University of Michigan · 2023 · XWTEBYE99LJU · [View certificate](#)